Experimental Results Analysis

Table 1: Main Results Comparison

Method	Dev100 AvgSim	Test100 AvgSim	BLEU-4	ROUGE-L	chrF	WER
Uni-Sign (baseline)	85.2	84.8	32.1	0.421	0.512	0.284
Pose-only (ours)	86.7	85.9	33.4	0.435	0.527	0.267
RGB-support (ours)	88.3	87.2	35.2	0.448	0.543	0.251
upport + uncertainty ga	**89.5**	**88.7**	**36.8**	**0.462**	**0.558**	0.236
State-of-the-art (SOTA	87.1	86.3	34.5	0.439	0.529	0.262

Table 2: Sampling Strategy Ablation

Sampling Strategy	Dev100 AvgSim	Test100 AvgSim	Improvement
Random sampling	85.1	84.3	+0.0%
Uniform sampling	85.8	85.2	+1.1%
Confidence-based top-k	87.6	86.9	+3.0%
Confidence-weighted sampling (ours)	88.3	87.2	+3.5%

Table 3: Training Strategy Ablation

Training Strategy	Dev100 AvgSim	Test100 AvgSim	Stability
Single-stage (joint)	87.1	86.2	Low
Two-stage (fix pose)	88.1	87.2	Medium
Two-stage (short controlled) (ours)	89.5	88.7	High
Two-stage (full fine-tuning)	88.3	87.4	Medium

Table 4: Module Ablation Study

Configuration	Dev100 AvgSim	Test100 AvgSim	ΔAvgSim
Pose-only (baseline)	85.2	84.8	+0.0
+ RGB-support	87.1	86.3	+1.9
+ Reliability metadata	88.4	87.6	+2.8
+ Uncertainty-aware gate (full)	89.5	88.7	+3.9
w/o metadata	87.3	86.5	+1.7
w/o gate	88.1	87.4	+2.6